More on Pivotal Quantities.

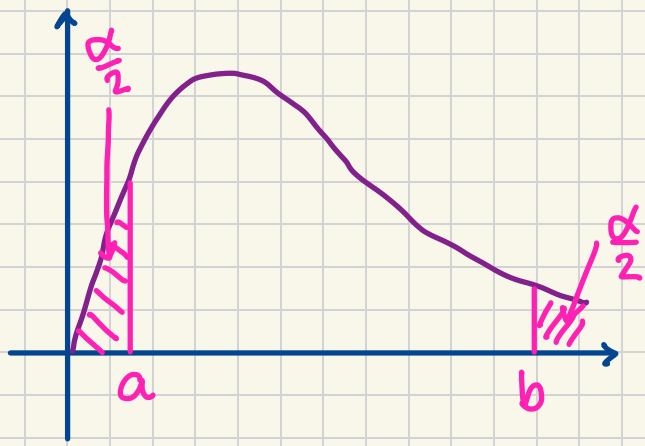
Example. $Y_1, \dots, Y_n$ from $E(\tau)$ w/ τ unknown

$$\bar{Y} \sim \frac{1}{n}\underbrace{(Y_1 + \dots + Y_n)}_{\Gamma(k=n,\ \tau)}$$

scale

$$\boxed{U = \frac{1}{\tau}\cdot\bar{Y}} = \frac{1}{\tau}\cdot\frac{1}{n}(Y_1 + \dots + Y_n) \sim \Gamma\left(n, \frac{\cancel{\tau}^{1}}{n\cdot\cancel{\tau}}\right) = \Gamma\left(n, \frac{1}{n}\right)$$

The dist'n doesn't depend on τ, so we do have a pivotal quantity.



Having picked a confidence level $C = 0.90$, i.e., $\alpha = 0.10$.

a = qgamma(0.05, shape = n, scale = 1/n) ←

b = qgamma(0.95, shape = n, scale = 1/n)

We know that

$$\mathbb{P}[a \le U \le b] = 0.90 = C$$

$$\mathbb{P}\left[a \le \frac{1}{\tau}\cdot\bar{Y} \le b\right] = 0.90$$

$$\mathbb{P}\left[\frac{1}{a} \ge \frac{\tau}{\bar{Y}} \ge \frac{1}{b}\right] = 0.90$$

$$\mathbb{P}\left[\underbrace{\boxed{\frac{\bar{Y}}{b}}}_{\hat{\theta}_L} \le \tau \le \underbrace{\boxed{\frac{\bar{Y}}{a}}}_{\hat{\theta}_R}\right] = 0.90 \qquad \square$$

Approximate Confidence Intervals for p

Consider a population in which a specific trait occurs w/ an unknown probability p.

Let $(Y_1, \dots, Y_n)$ be a random sample from the Bernoulli dist'n w/ the unknown p parameter.

Goal: Designing a confidence interval for p.

Idea: Look @ the natural point estimator for p

$$\bar{Y} = \frac{1}{n}\underbrace{(Y_1 + \dots + Y_n)}_{\sim B(n,p)}$$

Note: $\mathbb{E}[\bar{Y}] = p$

of course, the dist'n of $\bar{Y}$ depends on p.

We need a pivotal quantity.

By the deMoivre-Laplace Thm.

Set $S_n = Y_1 + \dots + Y_n$

Then, $\boxed{\dfrac{S_n - np}{\sqrt{np(1-p)}}} \xrightarrow{\mathcal{D}} N(0,1)$

Note: $\bar{Y} = \frac{1}{n} S_n$

$$\boxed{U = \frac{\bar{Y} - p}{\sqrt{\frac{p(1-p)}{n}}}} \xrightarrow{\mathcal{D}} \boxed{N(0,1)}$$

Still NOT a pivotal quantity.

But we can create an approximate confidence interval based on it!

Say, C is a confidence level.

Let

$$z^* = \Phi^{-1}\Big(\underbrace{1-\frac{\alpha}{2}}_{\frac{1+C}{2}}\Big) = \text{qnorm}\big((1+C)/2\big)$$

$$\mathbb{P}\left[\; -z^* \leq \mathcal{U} \leq z^* \;\right] \approx C$$

$$\mathbb{P}\left[\; -z^* \leq \frac{\bar{Y} - p}{\sqrt{\frac{p(1-p)}{n}}} \leq z^* \;\right] \approx C$$

$$\mathbb{P}\left[\; -z^* \sqrt{\frac{p(1-p)}{n}} \leq \bar{Y} - p \leq z^* \cdot \sqrt{\frac{p(1-p)}{n}} \;\right] \approx C$$

$$\mathbb{P}\Big[\; \underbrace{\bar{Y} - z^* \sqrt{\frac{p(1-p)}{n}}}_{\hat{\theta}_L} \leq p \leq \underbrace{\bar{Y} + z^* \cdot \sqrt{\frac{p(1-p)}{n}}}_{\hat{\theta}_R} \;\Big] \approx C$$

X :(

Usually, we write $\boxed{\bar{Y} = \hat{p}}$ and remember that $\hat{p}$ is a point estimator for p.

We construct our confidence interval for p as

$$\boxed{p = \hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

Problem 16.2. Gallup's inaugural measure of global loneliness shows over one in five people worldwide (23%) said they felt loneliness "a lot of the day yesterday."[1] *However, there were considerable variations between countries. For instance, out of 1000 individuals polled in Taiwan, 11% reported having felt loneliness "a lot of the day" before. What 90%-confidence would you report for the population proportion of Taiwanese who had felt lonely the day before?*

$$\boxed{p = \hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

$\longrightarrow$:

$\hat{p} = 0.11$

$C = 0.90 \Rightarrow$ $z^* = \Phi^{-1}\left(\frac{1+0.9}{2}\right) = \Phi^{-1}(0.95)$

$= 1.645$

$=$ qnorm(0.95)

$$p = 0.11 \pm \sqrt{\frac{0.11 \cdot 0.89}{100}}\,(1.645)$$

$\square$

[1] https://news.gallup.com/poll/646718/people-worldwide-feel-lonely-lot.aspx